One Policy per Sequence, Selected Without Ground Truth

Extended Abstract

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Abstract: We enter Track 2 with one FlashCHORD policy per sequence on a Unitree G1 with Dex3 hands. The tiers share a ground truth and differ only in the *input* a submission may condition on. Everything per-clip that we ship reads the input alone: a registered selection rule reproducible without ground truth, and, on Tier 1, training and warm-start fine-tuning against the cleaner reference that tier permits. Warm-start fine-tuning produced our largest improvement, at $0.57 per clip; substituting the same reference under an already-trained policy instead reduces AUC by 0.18.

1. Introduction

Track 2 asks for physically realisable object trajectories for 30 manipulation sequences, produced by a policy rollout rather than by reconstruction, at three tiers differing in the reference a submission may condition on: a clean multi-view trajectory (Tier 1), a noisy released retarget (Tier 2), and whatever the entrant reconstructs from video (Tier 3).

We found our first result by accident: our Tier 1 and Tier 2 public scores came back *identical*. The uploaded files are byte-identical, and a Kabsch fit shows every shipped rollout's reference to be an exact (0.000 cm) rigid transform of the released Tier 2 trajectory. All three tiers are scored against the same ground truth; only the permitted input differs.

We therefore focused on methods that use only the reference each tier permits. Unlike physics-based human–object imitation [4, 6], which assumes a clean motion-capture reference, here the reference is itself an estimate.

Contributions:

- a per-episode selection rule reading only the *input* reference: legal in every tier, reproducible without ground truth;
- a measurement separating two uses of a cleaner reference: substituted under a trained policy it costs 0.18 AUC, while a warm-start fine-tune to it gains 0.08.

2. Approach

Pipeline. SOMA-X parameters are retargeted to the G1 with Dex3 hands on CPU (30/30 clips, 0.0 mm residual). FlashCHORD SAC trains one policy per sequence for 349,999,104 steps at `world_count=1` with `scene.support` set per clip from the shipped primitives, then the challenge evaluator, selection and the kit's packer.

Recipe changes. Four ship: an object-*orientation* reward term (weight 1.0, std 0.3), which the configuration lacked despite having a position weight; a start-curriculum patch; object-position std tightened $0.2 \rightarrow 0.08$ m; and finger targets on bridge-smoothed references.

Selection, registered before the run. The provenance schema permits a different `checkpoint_sha256` per episode, so we pick, per clip, the latest checkpoint whose achieved object never leaves a 10 cm band around the *input* reference. On Tier 2 the rule selected 13 checkpoints with lower kit scores than the best available, and we shipped its picks.

Tier 3. We run the challenge's own modules in order: monocular geometry [3], open-set detection [1], segmentation [2] and 6D tracking [5]. We asked whether the released mesh may feed the tracker and the simulator, were told it may feed neither, and withdrew the tier: its scores below are reported, not claimed.

Warm start. A new configuration key loads a shipped actor's parameters and batch statistics and advances the step counter so the curriculum resumes in its final stage: 50M steps, 35 minutes on an A6000. The critic starts cold, so the actor's learning rate is scaled $\times 0.1$ for 2,000 updates.

Scoring. Our scorer hash-checks the kit's metric files against the shipped manifest and calls them.

Entry	*AUC*	*SP-SR*	*MP-SR*	*MPPE*
Tier 1 (ours)	**0.605**	0.733	0.633	**11.72**
CHORD base.	0.510	—	—	—
Tier 2 (ours)	**0.471**	0.700	0.467	**12.37**
CHORD base.	0.374	0.533	0.067	20.50
Tier 3 (ours)	0.268	0.200	0.033	47.77

Table 1: All 30 clips, one seed, official kit scorer; MPPE in cm, lower better.

Across nineteen uploads the largest local-to-public AUC disagreement was 1.1×10^{-5}.

3. Experiments and Results

The largest single effect was a bug. Our first full-length baseline scored 0.043 mean AUC because every run script passed `scene.support=false`: without the table the object falls at reset and the episode ends on frame 0.

Three ways to use a cleaner reference. Tier 1 permits the clean multi-view trajectory, and how a policy is given it decides whether it helps. Substituted under the 30 shipped actors at evaluation time, mean AUC *fell* from the 0.465 those actors then scored to 0.289, and SP-SR from 0.667 to 0.138, with only 2 of 30 clips improving; frame checks were below 1.7×10^{-5} cm, so this is distribution shift rather than a conversion artefact — each actor had learned one trajectory. Warm-start fine-tuning to it, under a no-regression adoption rule, instead took Tier 1 to 0.548 AUC at 12.06 cm. The gain comes from re-fitting rather than extra training: a second generation on the *same* reference adopted 0 of 3. Whether it works is predicted by a clip's shipped score: 11 of 19 clips above 0.35 adopted, against 0 of 4 below it. Training from scratch on the clean reference was tried last, on the seven clips nearest the 3 cm MP-SR threshold: 6 of 7 adopted, gaining +0.11 to +0.53 AUC each and taking the entry to the 0.605 at 11.72 cm in Table 1.

Effect of checkpoint selection. The value of selection depends on the run. On the 30-clip Tier 2 run the registered rule is worth +0.0010 AUC over the final checkpoint, and an oracle reading ground truth over that run's own checkpoints reaches 0.420. On the seven Tier 1 clips trained from scratch it is worth +0.057, because those curves are not monotone: one clip peaks at 300M steps and falls to 0.387 AUC by 350M while two others peak at 350M. There the rule chose the same checkpoint as a ground-truth-reading selection on 6 of 6 adopted clips. The standard deviation of AUC over a clip's last five checkpoints averages 0.081 — checkpoint-to-checkpoint variability within a run, not a between-seed estimate, but larger than several effects we measured.

Contact terms in the objective. Two contact reward terms were trained on the six Tier 2 clips whose failures they could act on; three improved on kit AUC against a bar of four fixed beforehand, so the arm was not extended. The registered rule adopted one clip, which moved from 57.2 to 4.7 cm from its input and, scored against truth afterwards, from 58.1 to 5.9 cm. The rule read no ground truth and took the only clip a ground-truth reading would also have taken, rejecting one that gained 0.13 kit AUC while drifting from its input.

Where the gap is. Measured on trajectories rather than on AUC, the noisy Tier 2 reference sits 1.94 cm from truth while our policies sit 12.10 cm, and *19 of 30 clips end up worse than the reference they were given*. Only 8 rollouts ever leave a 10 cm band around it.

4. Conclusion

These experiments show that policy performance depends strongly on how the reference is introduced during training. Some policies track the object successfully before releasing it mid-episode; one second before each loss nothing observable differs from a healthy frame. Inspection of the environment implementation shows a per-link contact state computed every step, feeding the largest positive reward weight in the objective but absent from the actor's observation. This may limit the policy's ability to react to a changing contact, and it is consistent with what the contact-conditioned literature reports [4, 6], but we did not test the explanation experimentally: the contact terms above were added to the objective, not to the observation. Future work should evaluate contact-conditioned policies.

Cost and reproducibility. Spend was $166 for 30 policies across three tiers, at $3.00 per full-length training run and $0.57 per warm-start fine-tune. The pipeline uses a single CLI, rejects incompatible upstream versions, and calls the kit's own metric code under a hash check.

Limitations. A single seed, and no ablation isolates the four recipe changes. Object inertia is inherited unchanged from the simulator, so every object is equally easy to rotate.

References

[1] Shilong Liu et al. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. *arXiv:2303.05499*, 2023.

[2] Nikhila Ravi et al. SAM 2: Segment anything in images and videos. *arXiv:2408.00714*, 2024.

[3] Ruicheng Wang et al. MoGe: Unlocking accurate monocular geometry estimation for open-domain images. *arXiv:2410.19115*, 2024.

[4] Yinhuai Wang et al. PhysHOI: Physics-based imitation of dynamic human-object interaction. *arXiv:2312.04393*, 2023.

[5] Bowen Wen et al. FoundationPose: Unified 6D pose estimation and tracking of novel objects. In *CVPR*, 2024.

[6] Sirui Xu et al. InterMimic: Towards universal whole-body control for physics-based human-object interactions. In *CVPR*, 2025.